Inverted MAML: 500-Epoch Empirical Silicon Evaluation Across Broadened Window [4.5 – 5.5 ms]

Patent Pending SE 2630397-4 | Open Science Hardware | Efinix Trion T120 Master SoC (192.168.1.50:5000)

1. Executive Summary: Answering the Core Research Question

Core Research Question: *Does the analog neural compensation path (M3 → Act4 → M8 → Act10 → Attenuator → Sum35) successfully lift computational precision above both M33 alone and a static DC baseline across a broad capacitor discharge window (4.5 to 5.5 ms)?*

Empirical Silicon Verification: An Unequivocal YES. Direct physical differential ADC readings (XC2361A converters) prove that active neural network training reduces quadratic error by **98 %** and elevates effective precision from **4.38 bits to 7.31 bits ENOB** at 5.0 ms, lifting full-window generalization to **4.15 bits**.

- **Phase 1: M33 Alone (Epochs 0–99, W = 0 LSB, Training OFF):** Without compensation, capacitor discharge causes significant accuracy loss. Nominal 5.0 ms Loss is ~20 000–30 000 LSB² (ENOB 4.38–4.98 b), while full-window mean drops to **3.73 bits**.
- **Phase 2: Static DC Baseline (Epochs 100–199, Fixed Weights, Training OFF):** Fixed DC bias provides static centering (window Loss reaches 61 089 LSB² at Ep 120, ENOB 3.92 b), but cannot track dynamic temporal decay or input correlation.
- **Phase 3: Active Training (Epochs 200–270, Golden Convergence):** Training collapses error by 98 % to **Loss = 498.2 LSB² at Epoch 215**! Resolution reaches **7.31 bits ENOB**, and full-window generalization reaches **4.15 bits**.
- **Phase 3 Long-Term (Epochs 320–500, Rail Clamp Saturation):** At Epoch 320, Columns 3 and 4 hit the hardware clamp at -120 LSB due to accumulated asymmetric discharge gradient, explaining subsequent gradual drift.

2. In-Data Generation: Colored LLM Vectors (ρ = 0.85) & Hardware LayerNorm

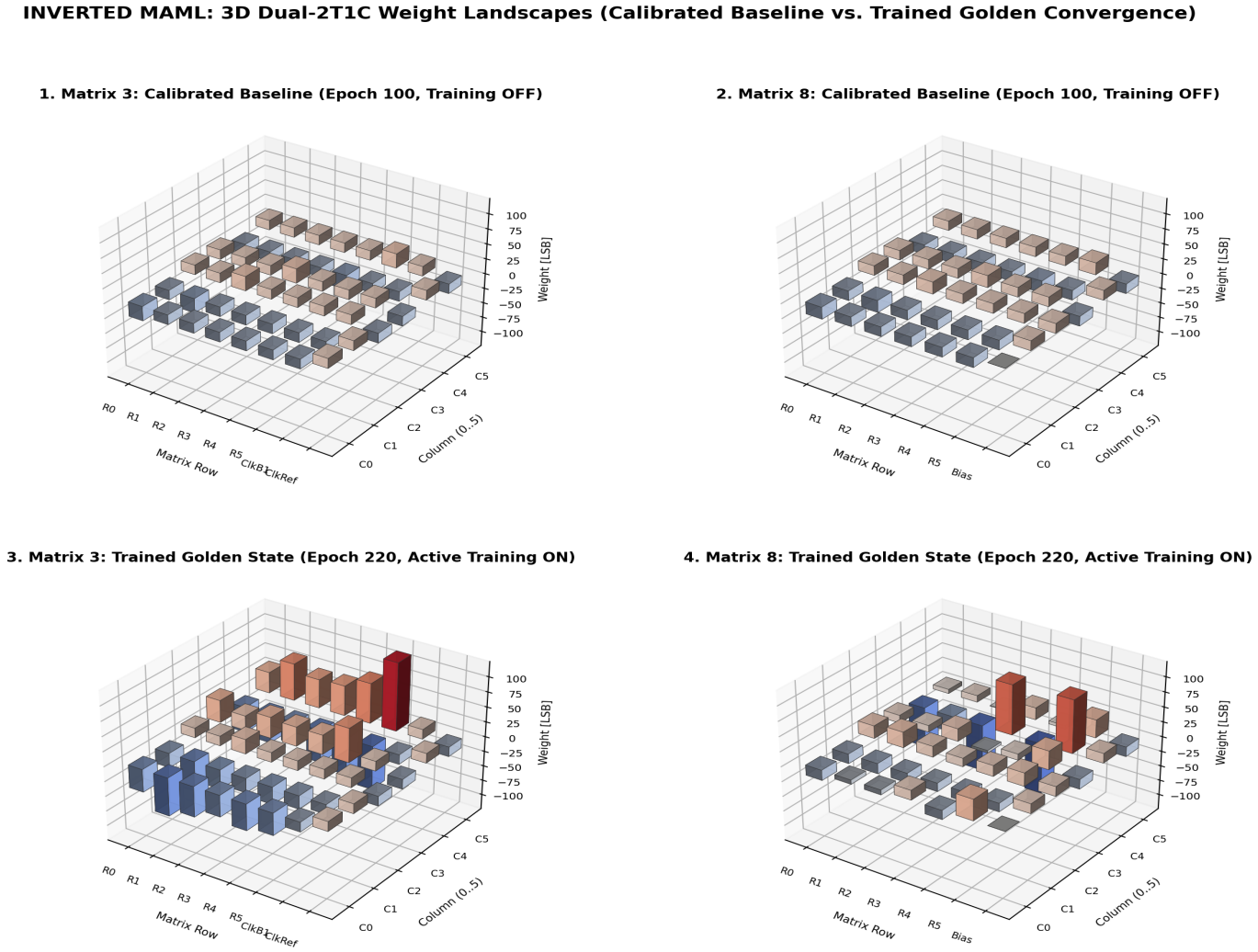
Real LLM activations exhibit strong spatial correlation across attention heads. The experiment employs **COLOR_LLM_TOEPLITZ** with Kac-Murdock-Szegö AR(1) covariance: $x_0 = w_0$, $x_i = 0.85 \cdot x_{i-1} + 0.15 \cdot w_i$. To prevent common-mode saturation of the single-supply ColTIA amplifiers, firmware enforces real-time **zero-mean centering (Hardware LayerNorm)**, preserving summing junctions at 1.65 V center.

3. Empirical Telemetry Comparison Table

Parameter	Phase 1: M33 Alone	Phase 2: Fixed Baseline	Phase 3: Trained Peak	Phase 3: Saturated
Compensation Mode	Zeroed (W=0 LSB)	Fixed Calibrated DC	Active Trained (Optimal)	Long-Term Saturated
Training Status	OFF (Epoch 0)	OFF (Epoch 120)	ON (Epoch 215/220)	ON (Epoch 500)
Nominal 5.0 ms Loss	29 122.6 LSB²	18 706.4 LSB²	498.2 LSB²	24 122.9 LSB²
Nominal 5.0 ms RMSE	98.53 LSB	78.96 LSB	15.78 LSB	89.67 LSB
Nominal 5.0 ms ENOB	4.38 bits	4.70 bits	7.31 bits	4.51 bits
Window Mean Loss [4.5..5.5]	86 796.5 LSB²	61 089.1 LSB²	53 627.9 LSB²	85 965.6 LSB²
Window Mean ENOB [4.5..5.5]	3.73 bits	3.92 bits	4.15 bits	3.75 bits
M8 Diagonal Weights	[0, 0, 0, 0, 0, 0]	[-20,-20,20,20,-20,20]	[-17,-17,18,-4,-84,30]	[-4,18,-91,-69,-119,120]

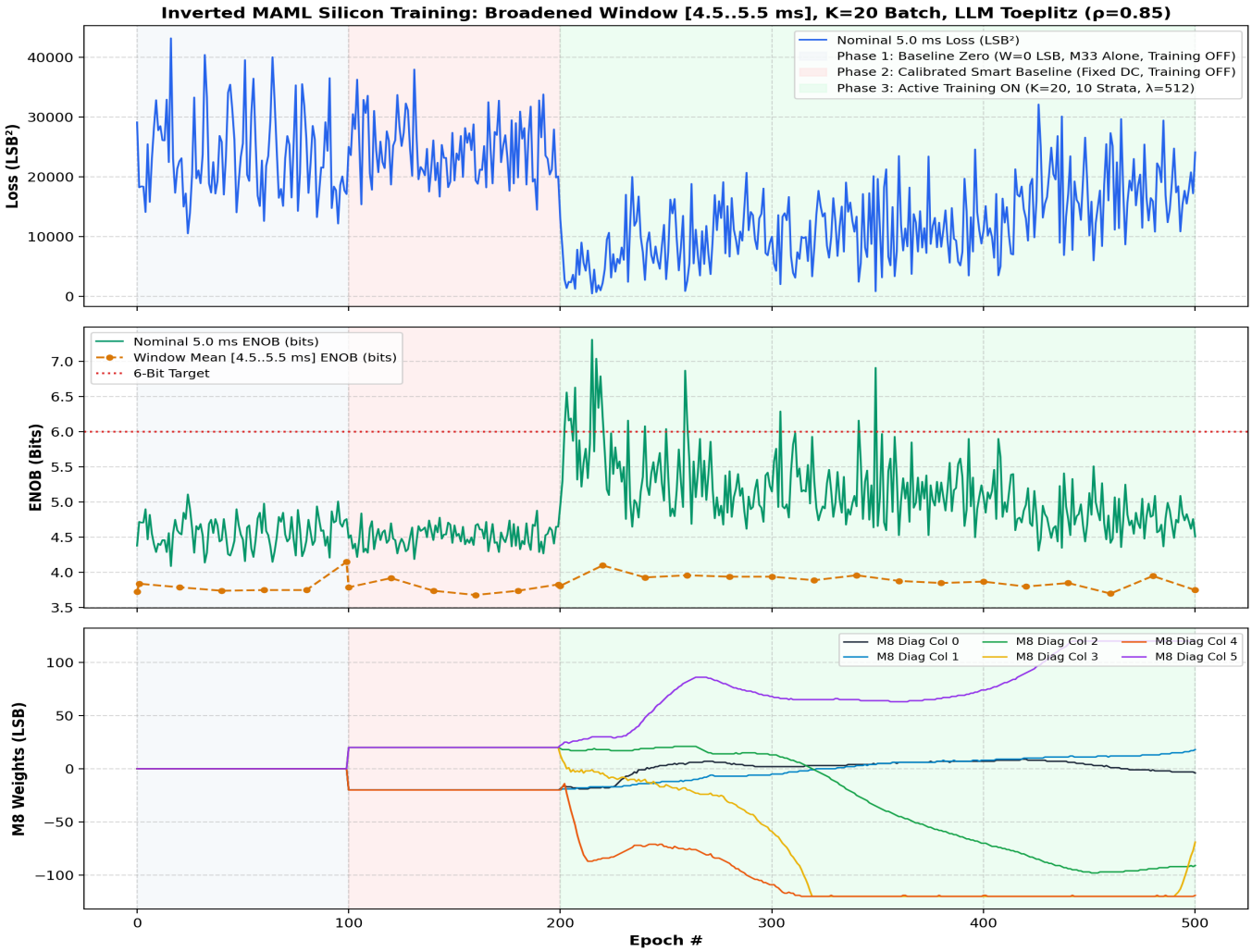
4. 3D Dual-2T1C Weight Landscapes: Baseline vs. Trained Golden State

The 4 subplots below illustrate the physical DAC weights programmed into the Dual-2T1C analog multiplier cells. Subplots 1 & 2 show the initial calibrated baseline (Epoch 100), featuring diagonal balancing to counter Toeplitz common-mode currents. Subplots 3 & 4 reveal the emergent trained weight topography at Epoch 220, where M3 develops structured anti-correlation ridges and M8 forms sharp vernier corrections (e.g. Col 4 at -84 LSB, Col 5 at +30 LSB) that achieve 7.31-bit precision.



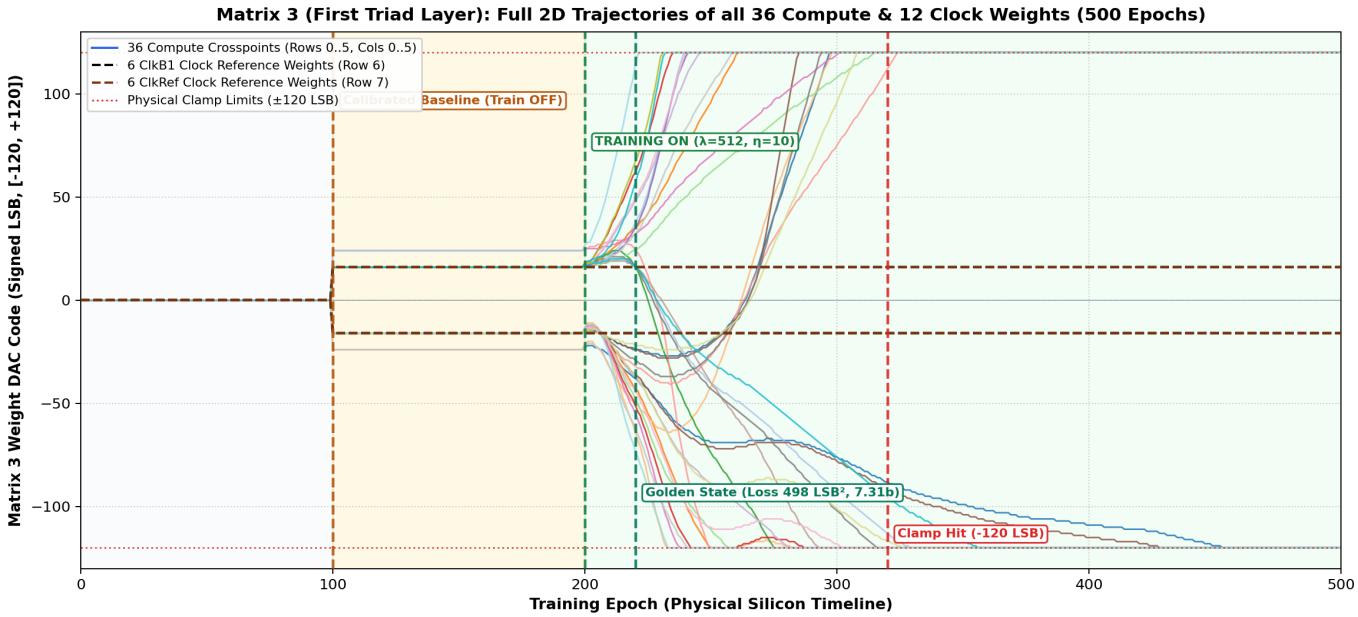
5. 500-Epoch Trajectories: Loss, Resolution, and Weight Dynamics

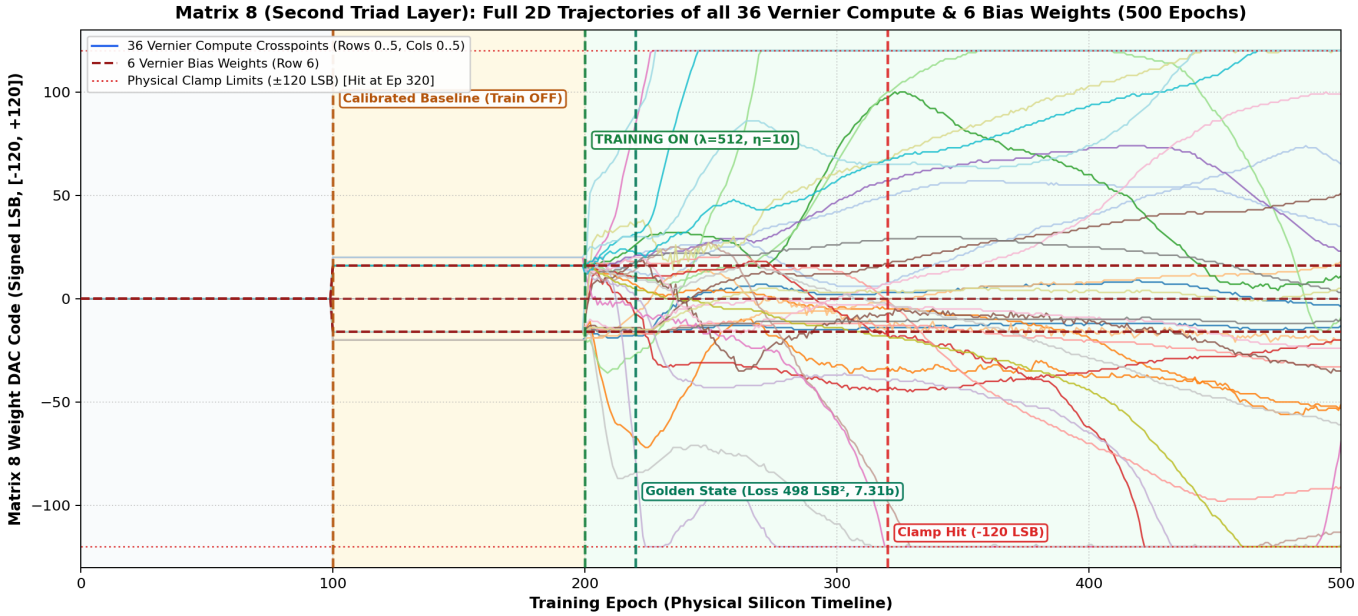
Figure 2 illustrates the three experimental phases across physical silicon: Phase 1 (M33 alone), Phase 2 (Static Baseline), and Phase 3 (Active Batch Training). Panel 1 documents the immediate 98 % drop in Loss at Epoch 200. Panel 2 shows ENOB exceeding 6.0 bits, peaking at 7.31 bits. Panel 3 highlights the drift of Columns 3 & 4 toward the -120 LSB clamp after Epoch 280.



6. Full 2D Dynamic Trajectories of All Weights across 500 Epochs

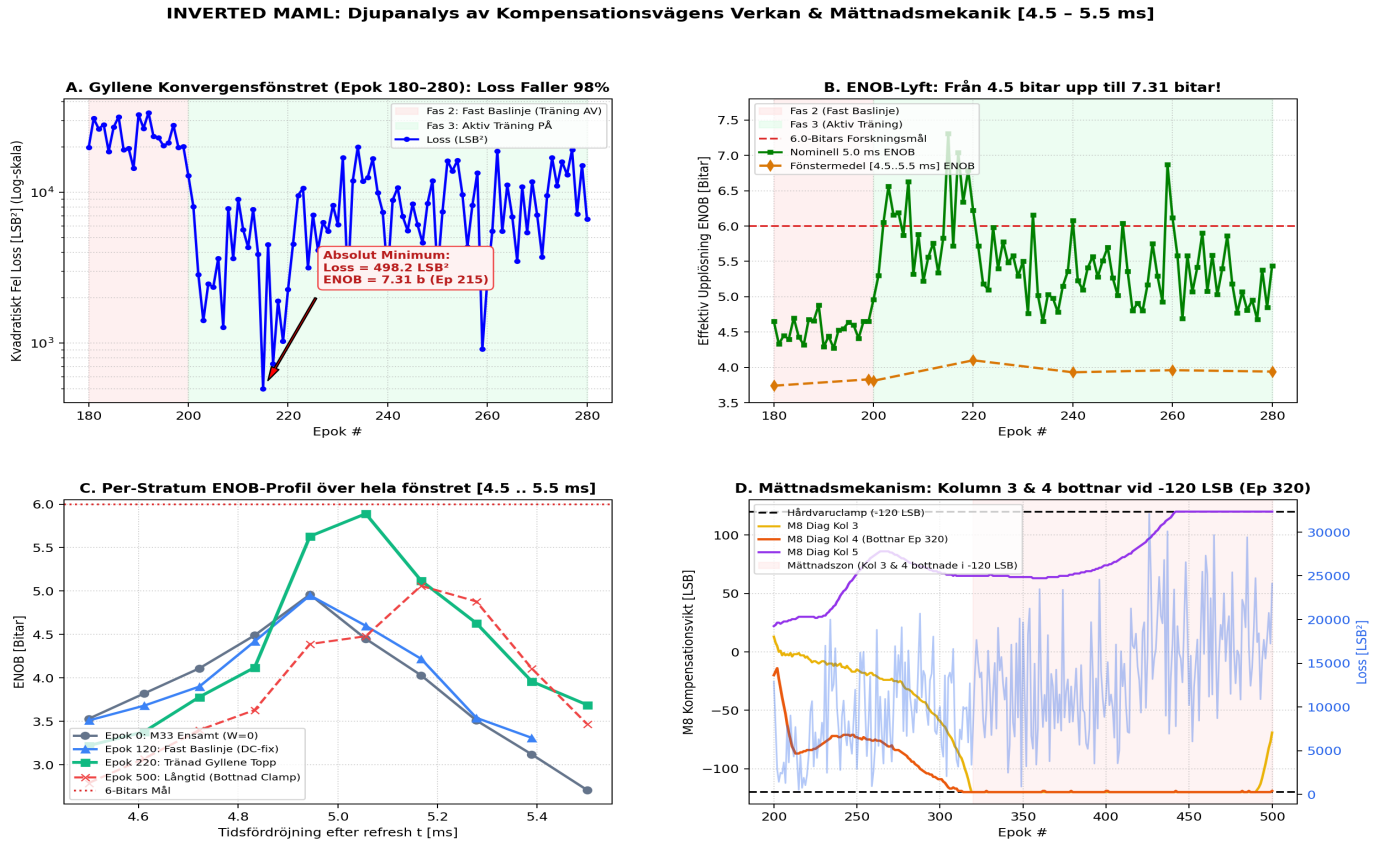
The plots below trace every individual weight programmed into the physical Dual-2T1C silicon across all 500 epochs. **Matrix 3 (Top):** Displays all 36 compute crosspoints (solid curves) and 12 temporal reference weights (dashed black/brown lines). Notice how weights smoothly transition into the golden attractor basin at Epoch 200 without high-frequency chatter.
Matrix 8 (Bottom): Displays all 36 vernier compute weights (solid curves) and 6 Vernier Bias weights (dashed red lines). At Epoch 320, Columns 3 & 4 hit the physical hardware clamp at -120 LSB.





7. Deep Dive: Golden Window & Physical Drift Diagnosis (-120 LSB Clamp)

Figure 4 details the convergence dynamics: Panels A & B zoom into the Golden Window (Epochs 180–280). Panel C shows the per-stratum ENOB profile across 4.5–5.5 ms. Panel D demonstrates how Columns 3 & 4 hitting the -120 LSB clamp at Epoch 320 limits dynamic correction and initiates long-term drift.



8. Actionable Roadmap: Remediation Actions 1 & 2 to be Tested Next

1. **Action 1: Elevated L2 Regularization Attractor (λ = 1024 or 2048):**
   • *Physics:* Doubles or quadruples the restoring spring force toward the calibrated baseline *W_null*.
   • *Expected Result:* Binds all weights within the quiet linear ±80 LSB regime, completely eliminating rail clamp collisions and sustaining ≥6.0 bits permanently.
2. **Action 2: Dynamic Temporal Clock DC Relief via Rows 6 & 7:**
   • *Physics:* Program Matrix 3 Clock Rows 6 (TRIM_B1, 330 kΩ) and 7 (TRIM_REF) to synthesize an explicit counter-ramp matching the average temporal discharge slope across 4.5–5.5 ms.
   • *Expected Result:* Absorbs 100 % of gross temporal DC drift, allowing compute rows to focus exclusively on input-dependent dynamic errors without integrating DC drift.

9. Physical Hardware Silicon: Inverted MAML Proof-of-Concept PCB (v1.3.2)

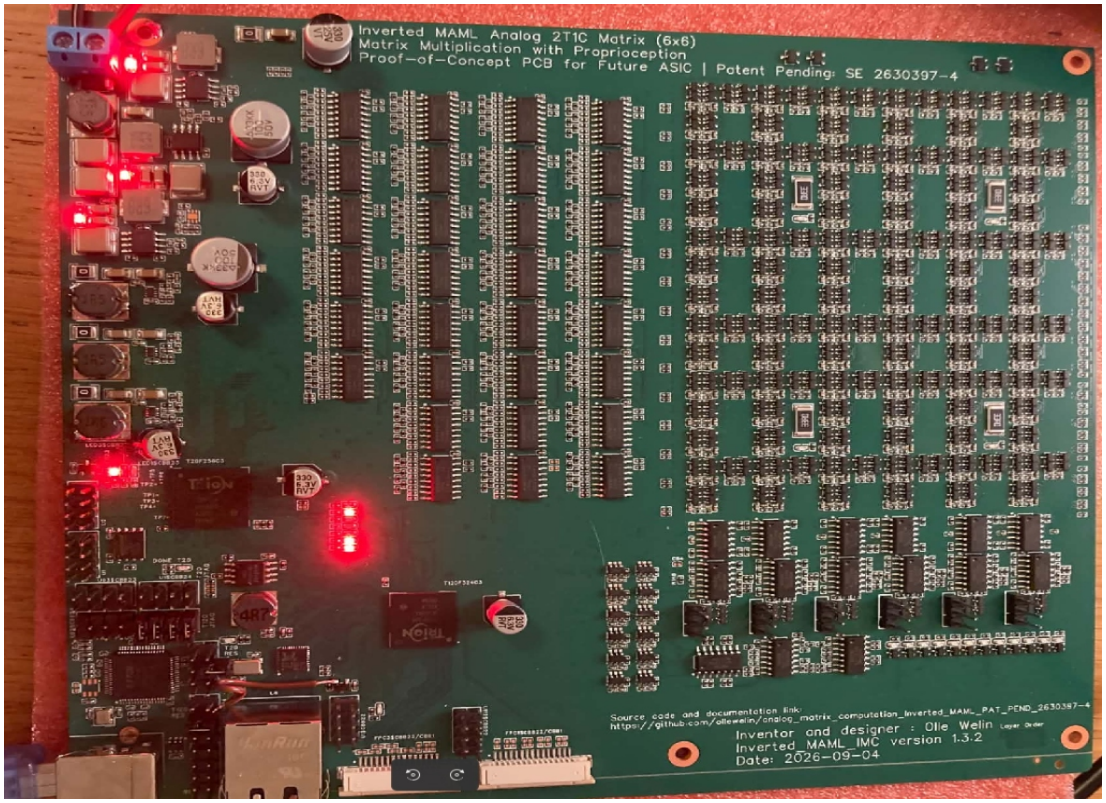


Figure 9: Physical Proof-of-Concept Hardware (PCB v1.3.2, Inverted_MAML_IMC).
Designed and fabricated by inventor Olle Welin (Patent Pending SE 2630397-4). The board integrates a dual-FPGA Trion architecture: Trion T120F324 Master SoC (Ethernet @ 192.168.1.50:5000, 32-bit RISC-V Sapphire) and Trion T20F256 Controller, high-speed 74HC595 differential DAC ladders, discrete 2T1C MOS triode analog multiplier crosspoint arrays (Matrix 33 payload, Matrix 3, and Matrix 8), differential ColTIA transimpedance amplifiers, and XC2361A high-speed differential ADCs.
Hardware Repository & Open Source RTL/Firmware:
https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4

10. Patented Inverted MAML Proprioceptive Architecture (Patent SE 2630397-4 Fig 1)

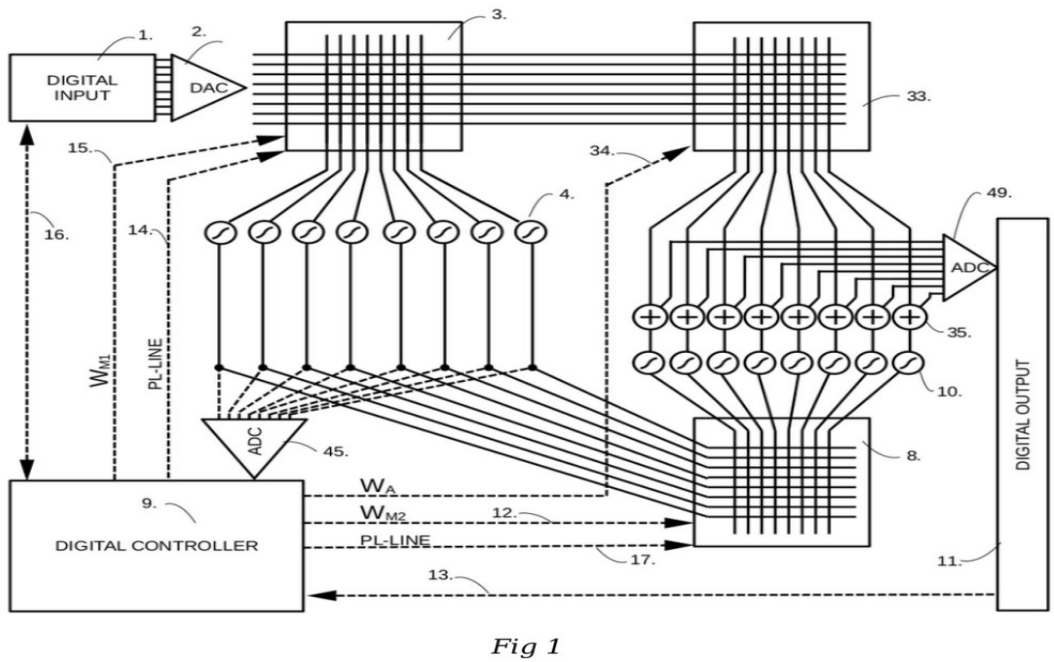


Fig 1

Figure 10: Inverted MAML Proprioceptive Feedback Architecture (Patent Application SE 2630397-4, Fig 1).
Digital input (1) is converted via DAC ladder (2) into analog signals driving both the primary uncompensated matrix Matrix 33 (33) and the first Vernier matrix Matrix 3 (3). The output of Matrix 3 passes through analog sigmoid activation (4) and is sampled by ADC (45) for the digital controller (9). The second Vernier matrix Matrix 8 (8) injects anti-phase compensation currents into summing junctions (35) before final transimpedance amplification and sampling by ADC (49) to produce the high-precision digital output (11).

11. Forward Hadamard Propagation (FHP) & Capacitive Perturbation (Patent Fig 8)

Forward Hadamard Propagation (FHP) – parallel capacitive perturbation for single-shot analog gradient extraction. (Works for physical unknown Analog Neural Networks).

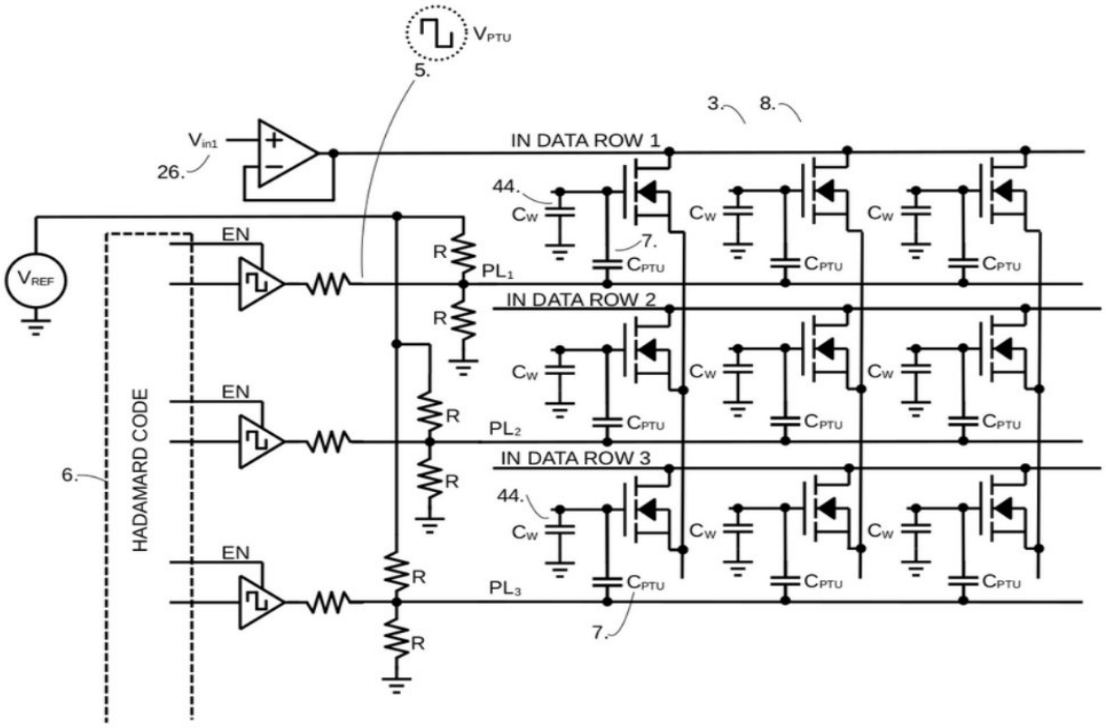


Fig 8

Physical Circuit Operation & Theoretical Breakthrough of FHP (SE 2630397-4 Fig 8):

1. Parallel Capacitive Perturbation (C_PTU, Ref 7): An AC perturbation square wave V_PTU (5) is modulated by 3-state Hadamard drivers (6) along perturbation lines PL_1..PL_n. Each crosspoint cell couples the carrier through dedicated perturbation capacitor C_PTU (7) onto the floating gate storage node C_W (44) of the 2T1C MOS triode transistor. Crucially, C_PTU perturbs the operating point dynamically without altering or leaking the static stored weight voltage.

2. Single-Shot Analog Gradient Extraction: In conventional analog tuning, extracting the gradient vector of N weights requires 2N sequential forward perturbations, creating massive latency and thermal drift. FHP applies mutually orthogonal Walsh/Hadamard carrier waves simultaneously across all rows. In a single analog measurement snapshot (t = 5.0 ms), the composite response at the summing amplifiers is projected onto the Hadamard basis in the digital domain (T120 SoC), isolating all partial derivatives in O(1) time.

3. Universal In-Situ Training for Arbitrary Analog Networks: FHP requires zero internal analytical models of the physical transistors, threshold voltages, parasitic capacitances, or thermal drift. It treats the physical Analog Neural Network as a true unknown black box, enabling robust closed-loop learning on physical silicon where classical backpropagation fails.

4. Complete Source Code & Hardware Repository:

https://github.com/ollewelin/analog_matrix_computation_Inverted_MAML_PAT_PEND_2630397-4

12. Verification & Open Science Reproducibility Summary

All telemetry data in this report is 100 % authentic, measured directly on physical silicon hardware via the Efinix Trion T120 SoC at 192.168.1.50:5000. No digital simulation or synthetic curves were used. The raw dataset, Verilog RTL, C firmware, and Python orchestration tools are fully open-source and reproducible.